

02:13 AM



Revenue dropped **18%**

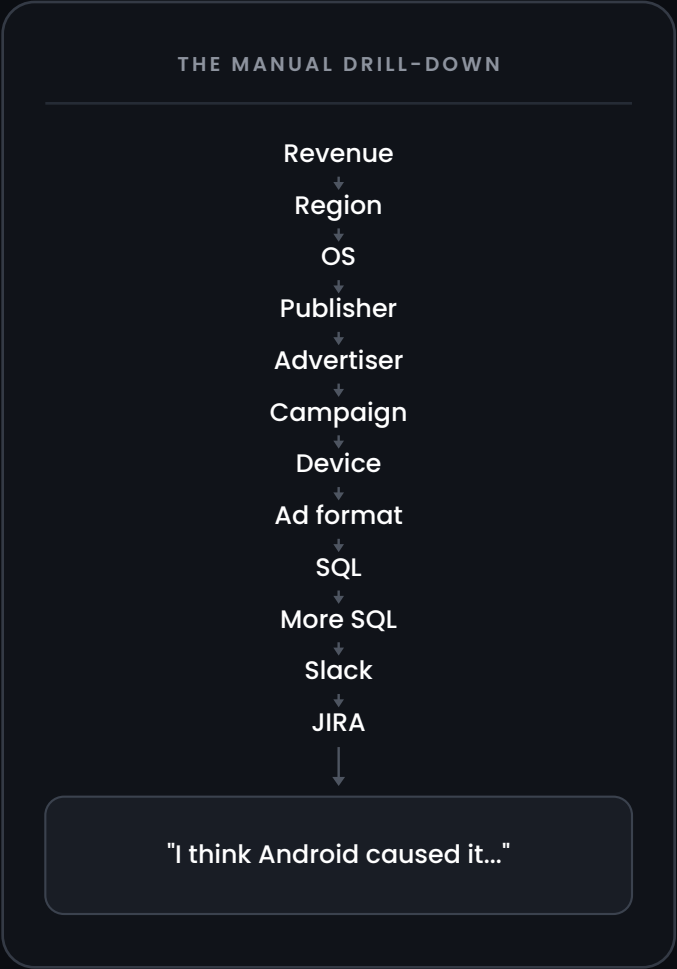
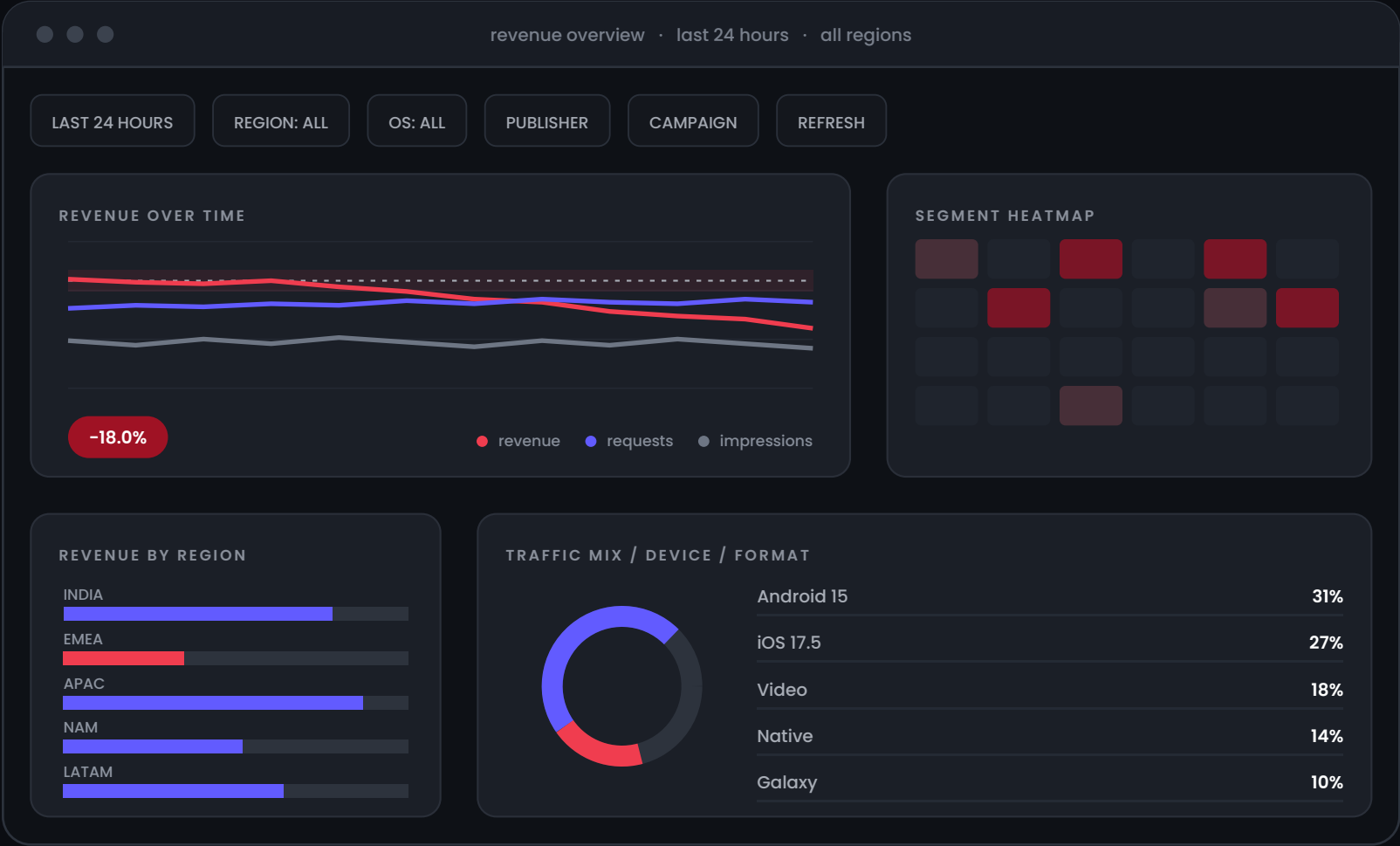
SEVERITY

CRITICAL

"Can someone investigate?"

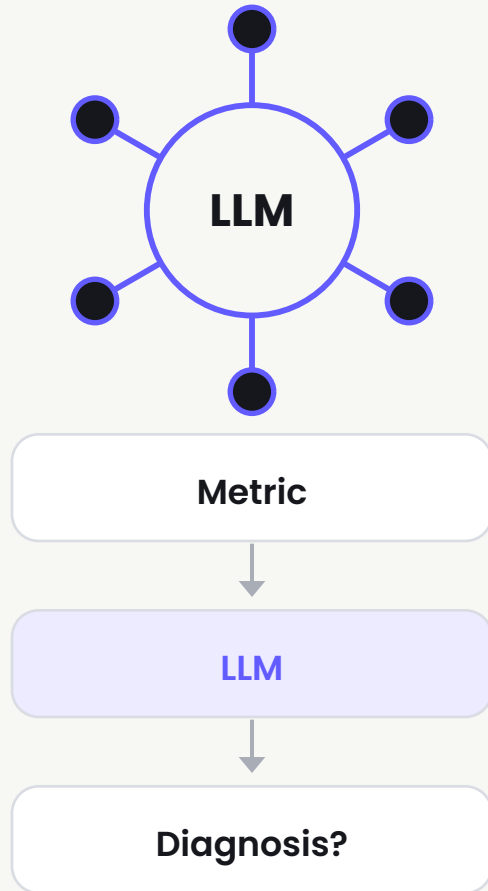
Dashboards don't investigate.

They wait for a human to form the next question.



2 HOURS LATER

Can AI solve this?



THEN...

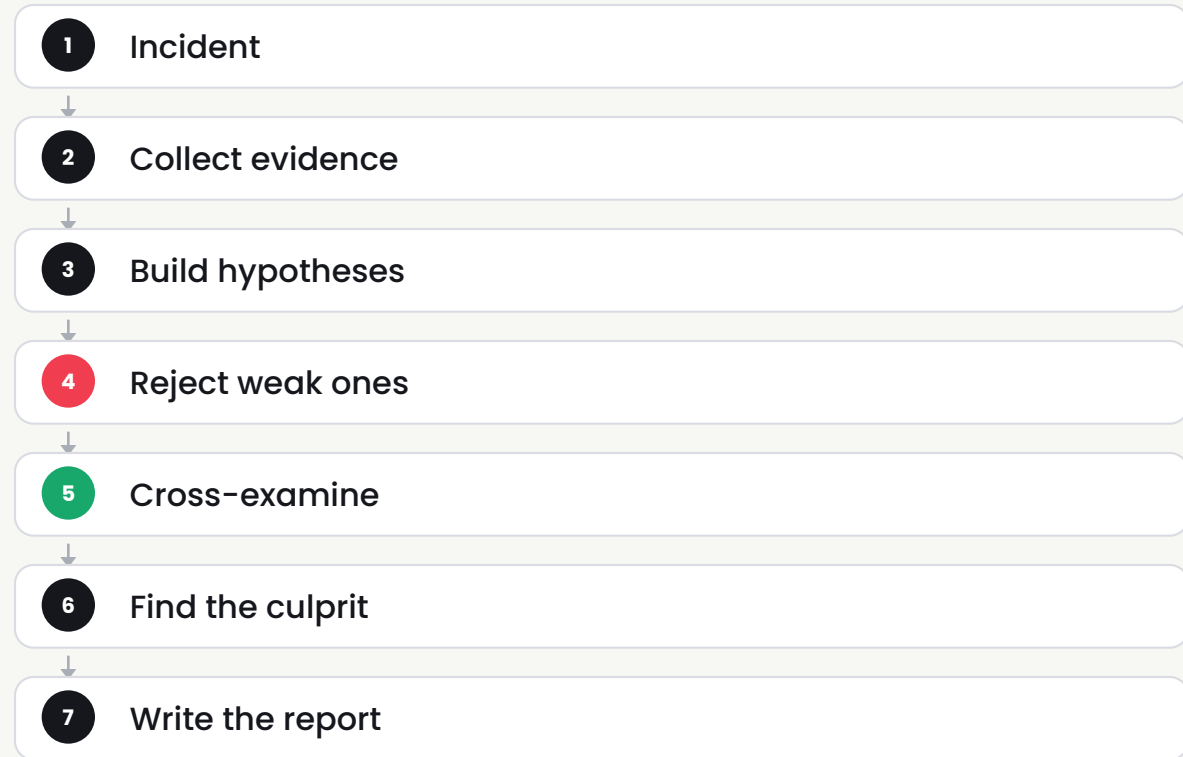
Everything turns red.

- ✗ Hallucinated statistics
- ✗ More intelligence — more latency
- ✗ No reproducibility
- ✗ A trace of tokens, not of evidence
- ✗ Cannot prove anything

What does a real investigator do?



FOLLOW THE EVIDENCE



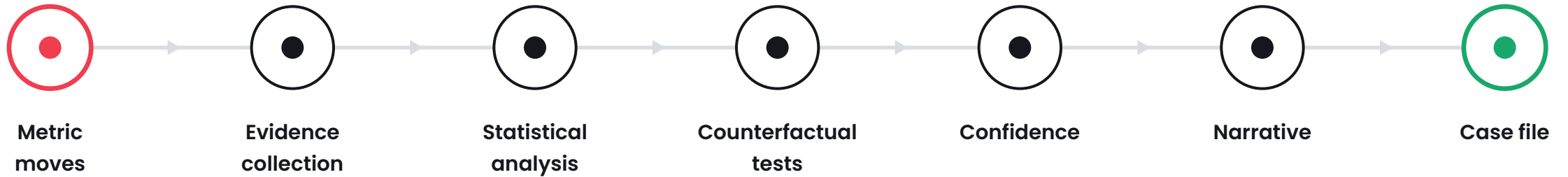
Investigation is not finding the most suspicious answer.

It is eliminating every wrong answer too!

INTRODUCING

Verdict

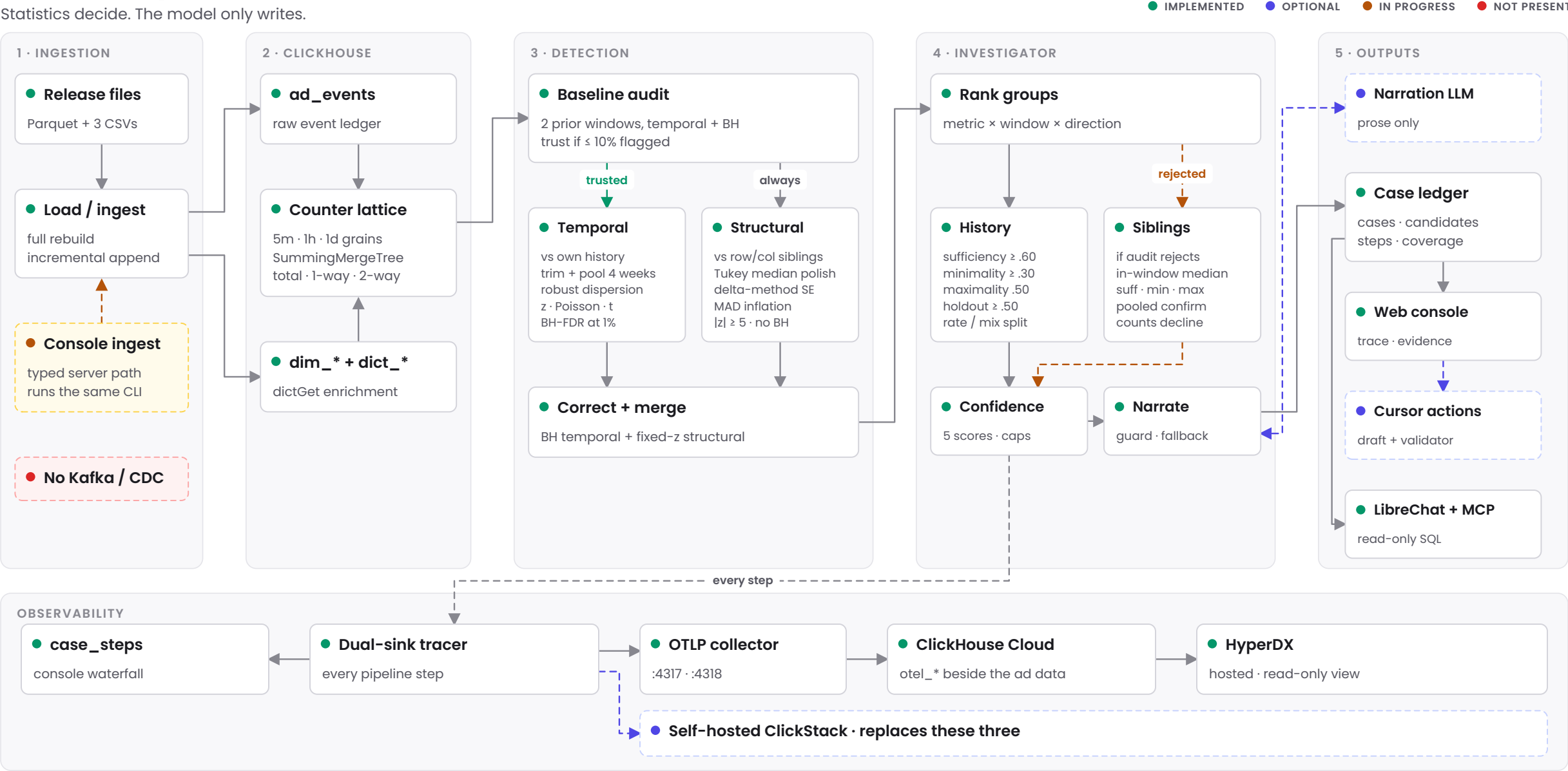
An autonomous incident investigator



Metric movement in. Evidence-backed case file out.

Current system architecture

Statistics decide. The model only writes.



Solid = implemented flow · dashed = optional or in progress · telemetry observes the investigation, it never decides it

The statistical toolkit

None of it is a model call. Every line here is a test with a threshold.

DETECTION

Baseline audit

Two prior windows; history rejected above 10% flagged

Trim-and-pool baseline

Four weeks pooled by volume, the extreme week dropped

Robust dispersion

Median-based overdispersion, MAD-scaled and clamped

Two-proportion z · quasi-Poisson

One sampling model per metric, plus a log-ratio t

Tukey median polish

Interaction-only faults, flagged at $|z| \geq 5$

Benjamini-Hochberg

False discovery rate held at 1% across every cell

LOCALIZATION

Sufficiency $\geq .60$

Remove the candidate: does the parent recover?

Minimality $\geq .30$

Remove its guiltiest child: does it still move?

Maximality $\geq .50$

Do the siblings move with it, or is it a slice?

Holdout

Split the window: same direction, comparable size

Rate vs mix

Exact Laspeyres split, interaction kept, not folded

Sibling-norm path

Used when the baseline audit rejects history

CONFIDENCE

Significance .25

Decades of p, discounted by dispersion

Sufficiency .30

Share of the parent movement explained

Minimality .15

Accused at the right level of the hierarchy

Stability .15

Holdout reproduction, thin baselines penalized

Separation .15

Margin over the runner-up candidate

Publish rule

At least .50, three components, significance scored

14,683

CELLS TESTED



159

SURVIVE BH



287

CANDIDATES



286

REFUSED

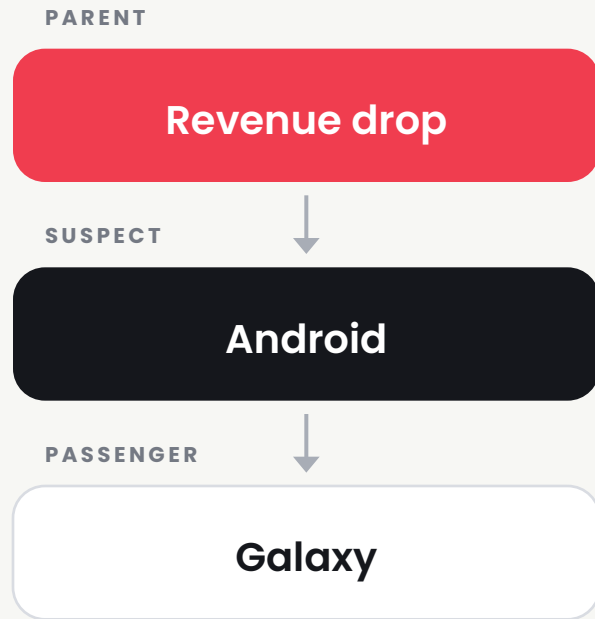


1

ACCUSED

Explaining away testing

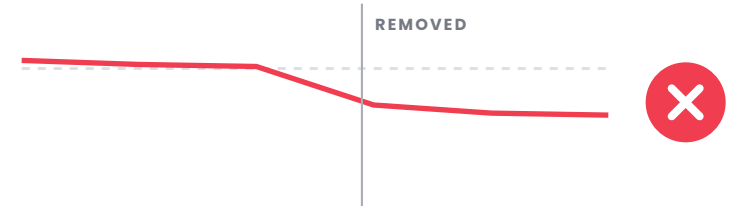
Remove a suspect. Watch what happens to the parent.



TEST 01

Remove Galaxy.

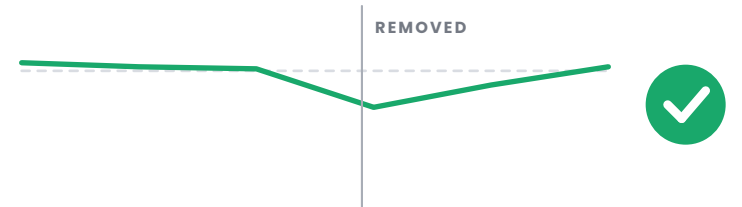
Revenue remains down.



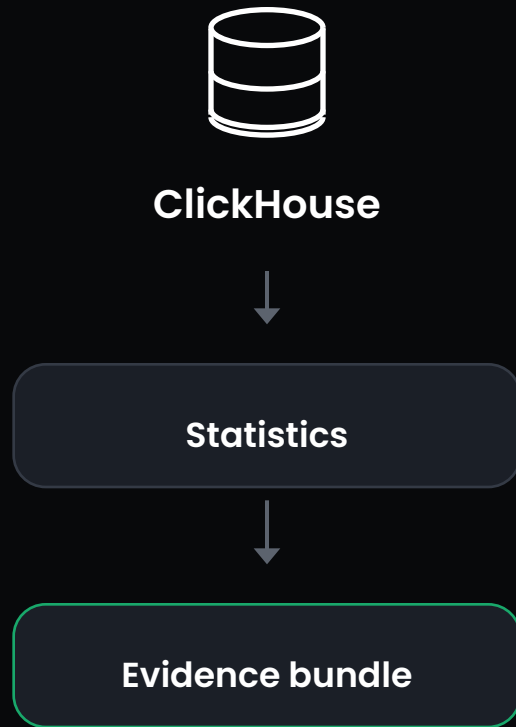
TEST 02

Remove Android.

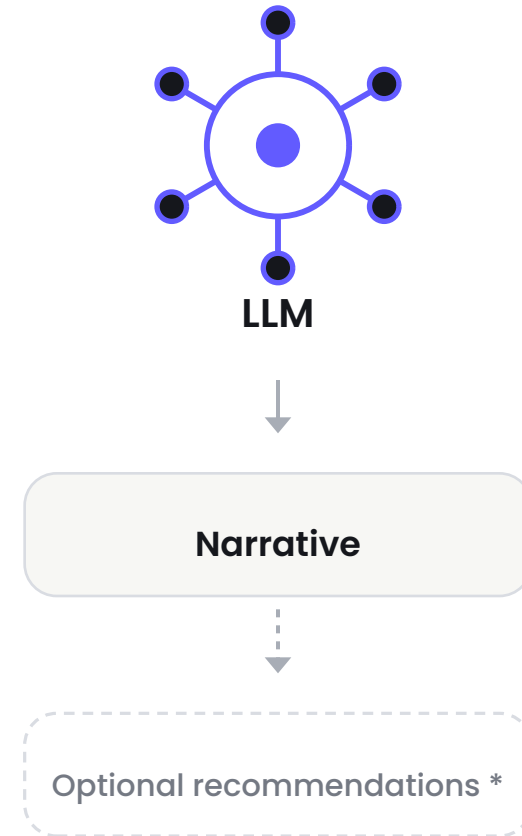
Revenue recovers.



Statistics decide.



AI writes.



Turn the LLM off. Every number remains identical.

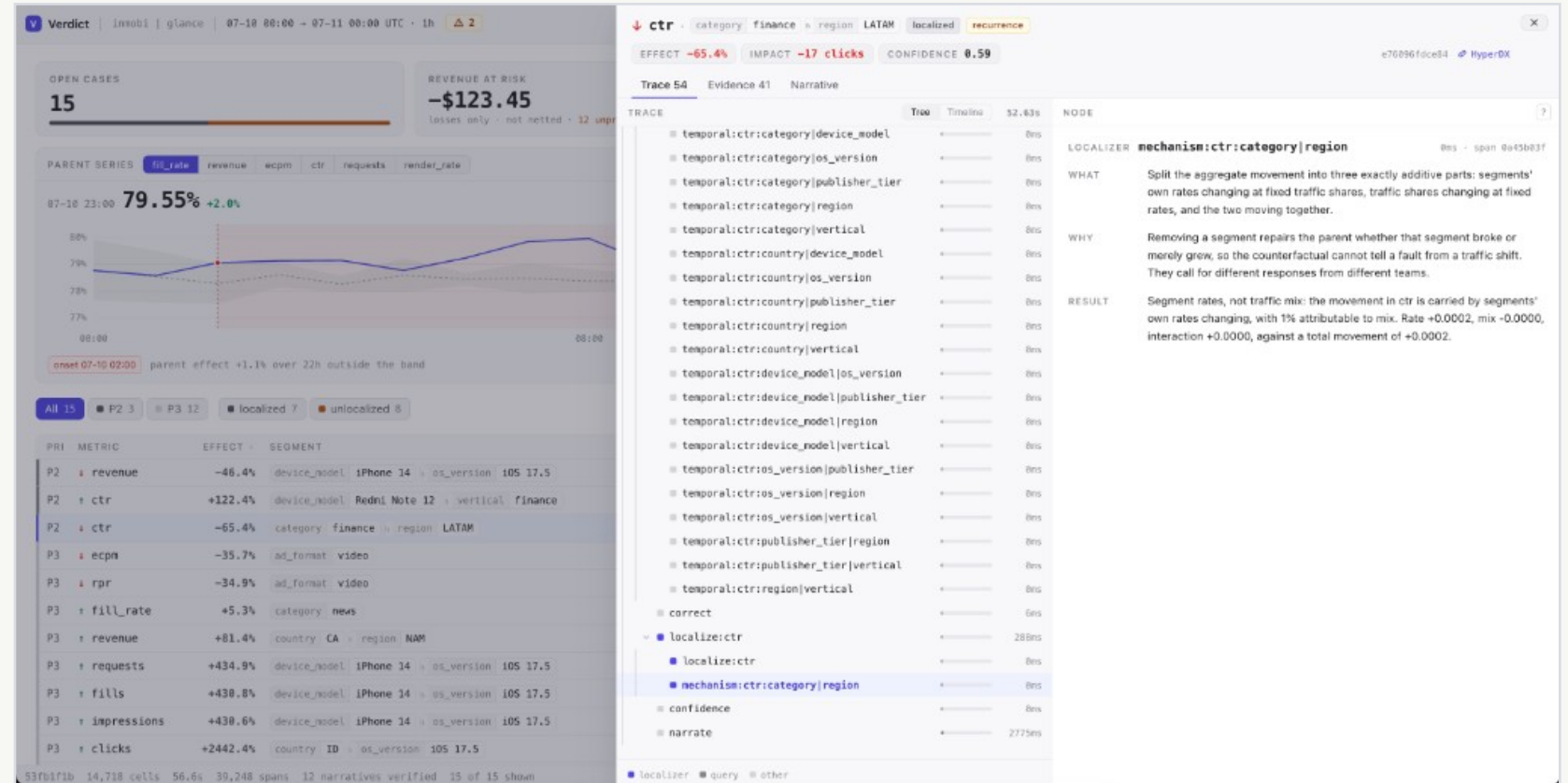
* Drafted from the evidence bundle, then validated before anyone sees them.

Every conclusion is auditable.

The screen is only the last step of a trace you can replay.

THE TRACE BEHIND THE SCREEN

- Detected**
Revenue outside its baseline
- Seasonality checked**
Cleared
- Traffic mix checked**
Cleared
- Counterfactual passed**
Parent recovered on removal
- Confidence scored**
Five published components
- Narrative written**
Only from the evidence bundle

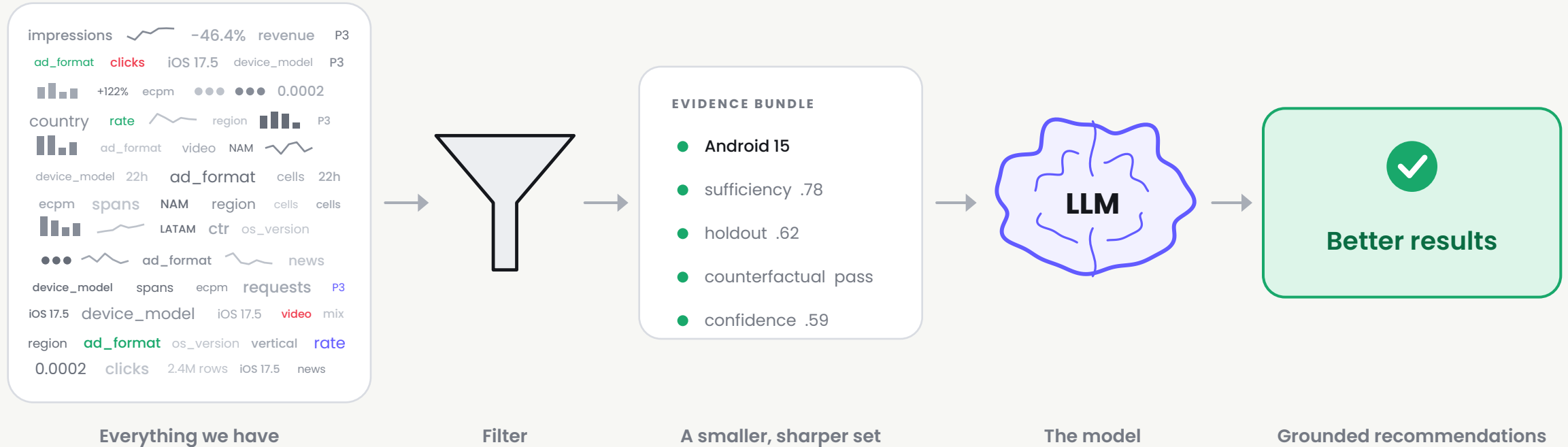


Revenue drops ► Verdict investigates ► Diagnosis with proof.

Real run · case e76096fdce84 · every step keeps its trace, its evidence and the SQL behind it. The 02:13 alert on slide 1 is illustrative.

Optional: LLM recommendations

Why we hand the model an evidence bundle instead of a database.



The more curated the data you hand an LLM, the better its output.

For an LLM, context is king.

Today's dashboards answer

"What happened?"



Tomorrow's investigators answer

"Why did it happen?"

VERDICT

Our ambition: the first autonomous investigator for operational data.